COMPUTER ENGINEERING
QUEZON CITY

Midterm Skills Exam

Course Code: CPE 212 **Course Title: Automating Server Management**

Intended Learning Outcomes Addressed

By the end of the lesson, students should be able to:

- Evaluate students ability to create and design a workflow that installs, configure and manage enterprise monitoring tools using Ansible as an Infrastructure as Code (IaC) tool.

Objectives

Create and design a workflow that installs, configure and manage enterprise monitoring tools using Ansible as an Infrastructure as Code (IaC) tool.

1. INSTRUCTIONS

1. Create a playbook that installs Prometheus in both Ubuntu and CentOS. Apply the concept of creating roles.
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
3. Show an output of the installed Prometheus for both Ubuntu and CentOS.
4. Make sure to create a new repository in GitHub for this activity.

Reminder: *Students are* ***NOT ALLOWED*** *to use any mobile device and AI throughout the skills exam. Students can search but with the "-ai" in their search string.*

2. MATERIALS AND EQUIPMENT

- VirtualBox
- 1 control node (Ubuntu), 2 managed nodes (Ubuntu and CentOS)
- Personal Computer

3. PROCEDURE

Sample output (students are allowed to add more steps as needed.)

Task 1: Prepare the Server

Task 2: Create the Prometheus User

Task 3: Create Prometheus Directories

Task 4: Download Prometheus

Task 5: Install the Prometheus Binaries

Task 6: Install the Prometheus Configuration Files

Task 7: Validate the Configuration
Task 8: Create the systemd Service
Task 9: Start and Enable Prometheus
Task 10: Verify Prometheus
Task 11: Access Prometheus Through a Browser

5. Conclusion